

How Renewable Energy Became One of the Biggest Business Opportunities

Cheap power, energy security and a race for the clean-tech supply chain have turned renewables from a climate policy into one of the largest capital allocation themes in energy.

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01 Why Cheap Energy Could Create the Next Superpower

The most powerful economies of the future will be the ones with the cheapest and most abundant energy, not the most oil.

About 20 percent

Share of world final energy consumption that is electricity

Three quarters

Share of global greenhouse gas emissions that comes from energy use

17.3 percent

Wind and solar share of world electricity in 2025, up from 4.5 percent in 2015

USD 2.2 trillion

Clean energy investment expected in 2026, against USD 1.2 trillion for fossil fuels

692 GW

Renewable capacity added worldwide in 2025, 85.6 percent of all new power capacity

Energy is bigger than electricity

Electricity powers lights, homes and data centres, but energy also means heat, cooling, steel, cement, chemicals and fuel for cars, trucks, ships and aircraft. Electricity is only about one fifth of the total.

In the early 1900s many cars were electric. The internal combustion engine displaced them and the world economy was rebuilt around oil, gas and coal. That system worked while nobody priced carbon.

Wind and solar have so far changed only the electricity part. At 17.3 percent of power they are still roughly 3.5 percent of total final energy (YD calculation). **The size of what is left to change is the size of the opportunity.**

Why carbon became a cost

From the late 1990s the evidence hardened that carbon in the atmosphere traps heat like a greenhouse. The costs are concrete and global: rising seas threaten coastal cities from Mumbai and Shanghai to Lagos and Miami, shifting rainfall disrupts the harvests that feed billions, and faster glacier melt brings floods first and weaker rivers later across Asia, Europe and the Americas.

The scientific consensus set 1.5 degrees Celsius of warming as the line beyond which changes become hard to reverse, and governments agreed to cut emissions. The problem was how, when entire economies ran on fossil fuels and coal plants could not simply be switched off.

From cost premium to cost advantage

At first wind and solar cost more than coal, and developing economies argued that the countries that caused the emissions should pay. Scale settled the argument: renewables became cheaper than new coal and gas in most markets, and affordability replaced climate as the reason to build.

World electricity demand rose 2.8 percent in 2025, faster than energy use as a whole, and China alone now accounts for one third of it. Data centres used about 1.5 percent of global power in 2024 and are set to double by 2030. **Whoever supplies that power cheaply, at scale and from domestic equipment, holds the advantage.**

How the reason to build renewables has changed



Sources: IEA, electricity and emissions data, Energy and AI and World Energy Investment 2026; Ember, Global Electricity Review 2026; IRENA, Renewable Capacity Statistics 2026. The 3.5 percent figure multiplies 17.3 percent by an electricity share of final energy of about 20 percent.

Why it matters for capital

Countries that remain energy dependent will sit in a weaker position on inflation, currency, industry and geopolitics. Cheap domestic electricity is becoming a national strategy, and national strategies attract capital.

02 The Future of Solar and Batteries

The sun is not the constraint. Capturing, storing and delivering its energy is.

Can the world run out of power?

No. The sun delivers more than 8,000 times as much energy to the Earth as the world uses at any moment (YD estimate). Even if solar output grew a hundredfold and met all of the world's energy needs, not only its electricity, the incoming resource would still be many times larger. The limits are capture, storage and distribution, and a power system built for a different century.

Technology keeps doing the heavy lifting

The global average cost of solar power has fallen 89 percent since 2010, and onshore wind 71 percent. A four-hour utility battery cost about USD 140 per kWh in 2025, down 30 percent in a single year and 95 percent since 2010. The cheapest solar markets, including India, China and Brazil, now produce at USD 35 to 37 per MWh. As both curves keep falling, solar stored for the evening is close to becoming the cheapest firm power available.

USD 44

Solar PV per MWh, 2025 global average

USD 33

Onshore wind per MWh, 2025 global average

USD 140

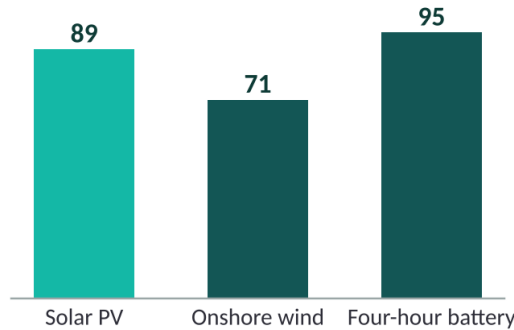
Four-hour battery per kWh, 2025

Too little power, or too much?

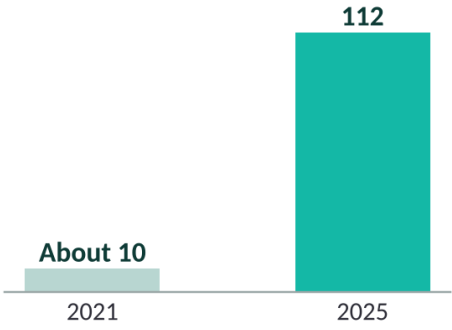
The common fear is that AI and electrification will force rationing between homes, industry and data centres. In many markets the visible problem is the opposite: surplus at midday. Between January and October 2025 wholesale prices fell below zero for 576 hours in Germany, 569 in Spain and 584 in the Netherlands, as solar output outran demand. At the same time about 2,500 GW of renewable, storage and large-load projects sit in grid connection queues worldwide. The shortage is in evening supply and in the lines that move power.

Sources: Ember; IEA, Electricity 2026; pv magazine on European negative prices (November 2025); US EIA; MNRE and PIB. Solar resource comparison uses incoming solar power of about 173,000 TW against world energy use of about 20 TW.

Cost decline since 2010, percent



Global battery storage added, GW



Sources: IRENA, Renewable Power Generation Costs in 2025; BloombergNEF via pv magazine (May 2026). Storage added 112 GW and 307 GWh in 2025, up 48 percent.

Solar will be everywhere, not only in solar parks

Generation is moving onto roofs, and in time onto building glass and vehicle surfaces as those technologies mature. The United States had about 59.5 GW of small-scale rooftop solar at the end of 2025, and India's rooftop scheme reached about 4 million homes and 12 GW within two years. Millions of households will both produce and use power, which changes how utilities plan, price and bill.

511 GW

Solar added worldwide in 2025, three quarters of all renewable additions

54 percent

China share of global storage additions in 2025, United States 16 percent

Grids have to catch up

Power must flow from wherever it is generated to wherever it is used, and almost every major market is short of the lines to do it.

03 Why America Has an Electricity Grid Problem

AI founders talk about power shortages because the US grid cannot move power where it is needed.

Four very different systems

FACTOR	UNITED STATES	EUROPE	CHINA	INDIA
Grid structure	Three separate interconnections with limited transfer between them	Continental synchronous grid run by national operators, linked by interconnectors	Ultra-high-voltage lines carry power from inland wind and solar bases to coastal cities	One synchronous national grid since 2013
Demand trend	Flat for a decade, now heading for a record in 2026 on data centres	Broadly flat, with industry under cost pressure	Grew 5.0 percent in 2025, one third of world demand	Planned at 6.41 percent a year to 2035-36
Data centre response	Behind-the-meter gas, nuclear and geothermal	Connection limits in crowded hubs such as Dublin and Amsterdam	Steered toward western provinces with surplus power	Grid supply plus contracted renewable power
Main bottleneck	Interregional transmission and permitting	Grid queues and midday surplus	Curtailment of remote wind and solar	Evening peak and transmission timing

Why the US is short of deliverable power

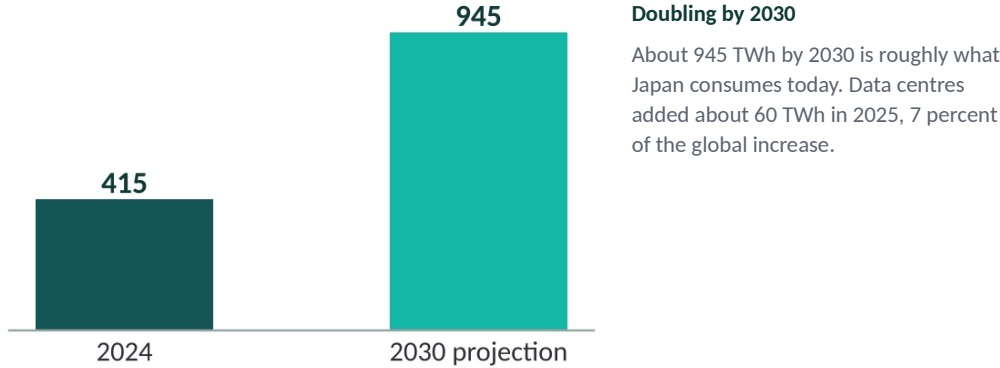
Wind-rich Texas and the Midwest cannot easily send power to California or New York. Planning lines across separate grids is slow, and building them across states with different political priorities is slower still. For years US demand did not grow, so new renewables simply replaced retiring plants. AI changed that. US consumption is set to rise about 2 percent in 2026 to a record 4,135 billion kWh, and data centre owners are building their own supply next to their sites.

Not every system is under the same strain

The United States holds about 45 percent of global data centre electricity use, China 25 percent and Europe 15 percent. Systems that already plan for steady demand growth, such as China and India, absorb data centres as one load among many. Mature systems with flat demand and fragmented grids feel the shock most.

Sources: IEA, Electricity 2026 and World Energy Investment 2026; Ember; US EIA; CEA projections for India. Characterisations of European and Chinese grids are YD summaries.

Global data centre electricity use, TWh



Sources: IEA, Energy and AI; Ember, Global Electricity Review 2026; US EIA, Short-Term Energy Outlook, September 2026.

The global grid task

About 2,500 GW of renewable, storage and large-load projects are waiting in grid connection queues worldwide, several times the capacity added in a year. The IEA estimates that better use of existing lines and flexible connection terms could release a large part of that queue, but most of it still needs new wires.

USD 550 billion

Global grid investment expected in 2026

2,500 GW

Projects stalled in grid queues worldwide

From a CFO perspective

Connection dates, curtailment and evening shortfalls are now revenue risks that belong in every project model.

04 How China Dominated the Clean Energy Supply Chain

It started 20 to 25 years ago with one decision: never again depend on the countries that export oil.

Four layers of control

1	End products	Western companies used China as a low-cost factory for panels, batteries and later EVs, helped by subsidies. Technology transferred, willingly or otherwise.
2	Capital equipment	China then built the machines that make cells, wafers and battery parts, so new factories elsewhere buy Chinese equipment.
3	Research	Work on the next generation of equipment and processes moved to China, so upgrades are designed there too.
4	Minerals and refining	Mines financed abroad, including cobalt in Congo, with refining kept at home. Subsidised scale priced other refiners out.

What China figured out that others did not

China saw its dependence on imported oil early. It built a large strategic oil reserve, which it has been drawing on during the 2026 disruption, and it set out to electrify its economy and own the equipment that electrification needs. It moved about a decade before anyone else, at a time when solar was only starting and most countries were not yet thinking about clean energy at all.

India, the United States and Europe each missed it for different reasons. India was inward looking and absorbed by domestic debate. The US had its own oil and gas. Europe assumed cheap Russian gas and Chinese factories would always be there. **China moved without a gun to its head. Every importing country now has one.**

One supplier, not ten	Oil can be bought from many countries, several of them allies. Solar equipment, battery materials and magnets largely come from one.
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Over 80 percent

China share of every solar manufacturing stage, heading to 95 percent for polysilicon and wafers

91 percent

China share of global rare earth refining, 2024; 94 percent of permanent magnets

74 percent

Congo share of world cobalt output, 2024. Chinese firms set to own over half by 2030

434 GW

Solar and wind China added in 2025, roughly two thirds of world renewable additions

Two kinds of countries

	FOSSIL FUEL EXPORTERS	FOSSIL FUEL IMPORTERS
Who	United States, Gulf producers, Russia and others	China, India, Europe, East Asia and much of Africa
Population	A minority of the world	About three quarters of the world's people
Interest	Keep a market for oil and gas	End dependence by becoming electrostates
Leader	United States, the largest oil and gas producer	China, 60.4 percent of capacity non-fossil

Others are now trying to follow. The United States restricts Chinese content in subsidised projects, the EU wants 40 percent of its clean-tech needs made at home by 2030, and India protects module and cell making from imports. The cost gap is real: the IEA puts Chinese solar manufacturing costs 10 percent below India, 20 percent below the United States and 35 percent below Europe.

Sources: IEA, Special Report on Solar PV Global Supply Chains and critical minerals commentary (October 2025); Benchmark Mineral Intelligence on cobalt; China NEA via pv magazine; IRENA; Ember, energy security report (2026). The two-thirds share is a YD calculation using IRENA global additions of 692 GW.

05 Why the US and Europe Fell Behind in Clean Energy

One chose oil and gas. The other outsourced its factories and its fuel.

United States: comfortable with fossil fuels

Many assume the Gulf or Russia leads world energy exports. The United States is now the world's largest producer of oil and gas and its largest LNG exporter, and US gas shipments to Europe rose from 18.9 bcm in 2021 to 45.1 bcm in 2024.

Policy swings with each administration. The 2022 Inflation Reduction Act put large incentives behind renewables. The current administration has doubled down on fossil fuels and formally left the Paris Agreement in January 2026. Higher global oil and gas prices hurt US drivers at the pump, but US producers and exporters gain, and on a net basis the country benefits. **For the US, supply disruption elsewhere is a commercial opportunity.**

Europe: committed to climate, exposed on energy

Europe moved much of its manufacturing to China and relied on cheap Russian gas, which supplied 45 percent of EU gas imports in 2021. After the invasion of Ukraine it cut that to 19 percent by 2024 and turned to US LNG, a relationship that now depends on transatlantic politics. High energy prices are eroding the competitiveness of European industry, and higher labour and power costs mean Europe cannot realistically become a clean-tech manufacturing base.

Europe has held to its climate commitments. Since 1 January 2026 its Carbon Border Adjustment Mechanism has priced the carbon in imported steel, aluminium, cement, fertilisers, hydrogen and electricity. That protects its climate goals but makes it a higher-cost economy, and decisions across 27 member states are slow. US GDP per head is about 34 percent above the EU on a purchasing power basis, mostly a productivity gap.

45 to 19 percent

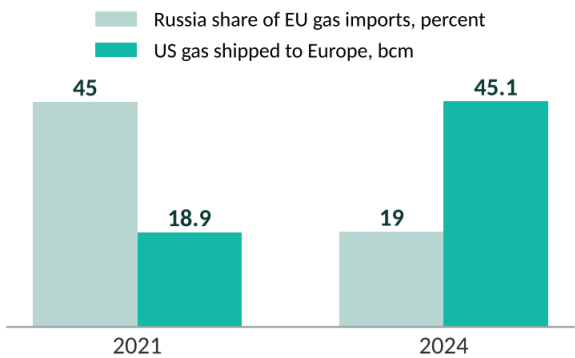
Russia share of EU gas imports, 2021 to 2024

34 percent

US GDP per head above the EU, purchasing power basis

Opinions on strategy and investor implications are YD views.

Europe's gas supply, 2021 and 2024



Reading the chart

Russia's share of EU gas imports fell from 45 to 19 percent between 2021 and 2024, while US gas shipped to Europe more than doubled. Europe reduced one dependence by deepening another.

Sources: Al Jazeera citing EU data (October 2025); US EIA; European Commission; ITIF (November 2025); US Congress Research Service on Paris withdrawal.

Five strategies compared

	STRATEGY	MAIN WEAKNESS
United States	Export fossil fuels, protect the home market	Policy reverses with each election
Europe	Price carbon at the border, buy gas from allies	High energy and labour costs
China	Electrify and own the clean-tech supply chain	Trade defences against its exports
India	Build renewables fast, climb the supply chain	Late start, higher cost of capital
Gulf producers	Export hydrocarbons, build solar at home	Export route exposed, as 2026 showed

What it means for investors

Clean-tech manufacturing capital will not flow to Europe at scale. It will flow to countries with cheap power, large home markets and policy protection, which today points to China, increasingly India and, for some technologies, the Gulf.

06 Can Energy Wars Accelerate the Renewable Revolution?

Yes. Every supply shock teaches importers that the safest supply chain is one inside their own borders.

2026 in five dates

DATE	EVENT
28 February	Military action begins. Most shipping stops transiting the Strait of Hormuz, which carried about one fifth of global oil and LNG
11 March	IEA members agree the largest emergency stock release in their history, 400 million barrels
31 March	Brent ends the quarter at USD 118 per barrel, up from USD 72 in late February
August	Extra fossil import costs since March estimated at USD 35.5 billion for China and USD 22 billion for India
28 September	Brent above USD 107 as a proposal to reopen the strait is rejected

Who wins in the short run, and who wins later

When Gulf supply is cut, buyers turn to whoever has spare supply, as Europe did with US gas after Russia and as Asian buyers did for LPG and LNG in 2026. Some industries, such as fertilisers, cannot run without gas, so they pay whatever it costs. In the short run the exporters win.

In the long run the effect reverses. Importers pay more, see how fragile long supply chains are, and invest so they need less imported fuel: electrifying their economies, building domestic supply chains and moving industry and transport to electricity. **The more energy is used as a weapon, the faster importers electrify.**

USD 1.7 trillion

World spending on fossil fuel imports, 2024

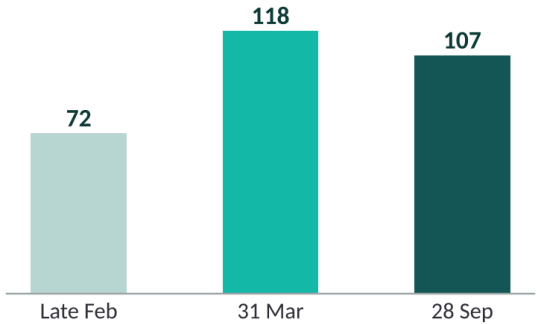
USD 160 billion

Added annual import cost for every USD 10 rise in oil

1.7 million

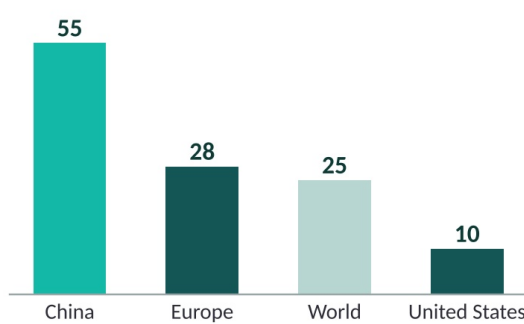
Barrels a day of oil displaced by EVs in 2025

Brent, USD per barrel, 2026



Sources: US EIA; IEA, Global EV Outlook 2026; Al Jazeera (28 September 2026); CREA via Business Today (August 2026); Ember, energy security report (2026).

Electric share of new car sales, 2025, percent



From climate, to cost, to security

Climate change drove the first phase of renewables but always felt distant to most people. Affordability drove the second. Energy security drives the third, and it is felt immediately: in the fuel bill, in the currency, in inflation and in a country's bargaining position. About three quarters of the world's people live in countries that import fossil fuels.

How importers and exporters are responding

REGION	RESPONSE
China	Draws on large oil stocks while EVs passed half of new car sales in 2025
Europe	Cut Russian gas from 45 to 19 percent of imports and added 77 GW of renewables in 2025
India	Record 55.3 GW of non-fossil capacity added in FY2025-26, with domestic module making
Gulf exporters	Building solar at home, from 24 GW today toward a GCC target of 165 GW by 2030

Sources: IEA; IRENA; MNRE; AGBI citing IRENA and the Baker Institute (June 2026).

07 Why Investing in Research and Development Matters

Countries that only install technology designed elsewhere capture the least of its value.

The research race

Research intensity separates the countries that set the technology agenda from those that buy it. Korea spends about 5 percent of GDP on research and development, the United States and Japan about 3.4 percent and China about 2.6 percent, against an OECD average of 2.7 percent. India spends 0.64 percent, and most emerging markets less. Few companies anywhere plan their green industrial strategy 15 to 20 years out.

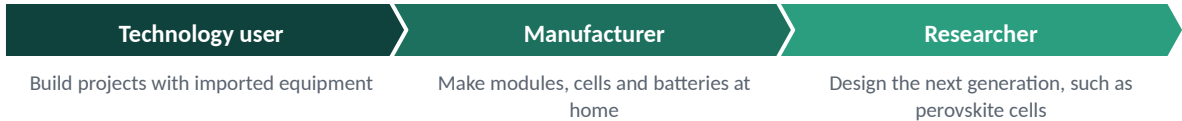
Why companies underinvest

Capital markets reward the quarter. Investors in most markets judge companies on quarterly results and give little credit for research spending that pays off in ten years.

Survival comes first. Clean energy developers spend their early years proving they can build and then fighting for share in fiercely competitive markets, which leaves little room to plan the next generation of technology.

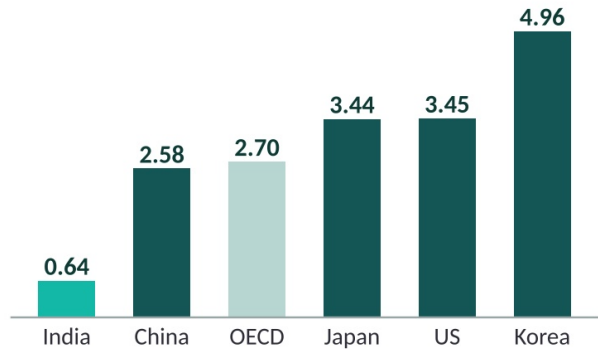
Research and industry rarely meet. Outside a few leading economies, university research is often academic in focus and links between institutes and companies are weak.

The ladder countries and companies climb



Most markets are on the first rung. India, the United States and Europe are rebuilding the second with trade protection and incentives. China, Japan, Korea and the United States lead the third. The same applies to people: companies that develop talent internally build the long-term muscle that research needs.

Research and development spending, percent of GDP



Sources: OECD data for 2023 via SSTI; Economic Survey 2025-26 for India.

Reading the chart

Research intensity in Korea is nearly eight times India's. On larger economies, the absolute gaps with China and the United States are wider still.

China's share of each step of the solar value chain



Others are working backwards from the module. The EU Net-Zero Industry Act aims for 40 percent of its clean-tech needs to be made in Europe by 2030. The United States restricts Chinese content in subsidised projects from 2026. India has over 200 GW of listed module capacity but only about 31 GW of cells. Batteries, motors and electrolyzers need the same climb.

A shared effort

Governments set the policy, companies invest in research and supply chains, and consumers choose electric vehicles, rooftop solar and batteries.

Sources: IEA, Special Report on Solar PV Global Supply Chains; European Commission; Kirkland and Ellis on the 2025 US tax law; MNRE and Down To Earth (2026). Ladder placements are YD views.

08 Will the World Move Away From Fossil Fuels, and Why It Must

Not quickly. Long-life assets lock in today's choices for 30 to 40 years.

Will the world ever move away from fossil fuels?

Not soon, and not completely for decades. Fossil fuels still produced about 57 percent of the world's electricity in 2025, although renewables overtook coal for the first time. Plants already built will run until the end of their lives, and grids still need steady, controllable output until storage and other flexible supply can replace it.

SECTOR	WHY CHANGE IS SLOW	WHAT WOULD SPEED IT UP
Coal power	Plants run to the end of their lives and provide grid stability	Storage, flexible supply and retirement incentives
Road transport	Hundreds of millions of petrol and diesel vehicles on the road	EV incentives, charging networks, lower running costs
Aviation	Sustainable aviation fuel is scarce and expensive	Mandates and scale in fuel production
Shipping	A new ship takes four to five years to deliver and runs for decades on its chosen fuel	Orders for green methanol and ammonia ships
Steel	About 70 percent of world steel comes from coal-based blast furnaces that run for decades	Electric arc furnaces and green hydrogen
All sectors	No company changes without pressure or reward	Policy, carbon pricing, or clean options becoming cheaper

The lock-in test

Every long-life asset approved this decade, whether a steel plant, a ship or a power station, either locks in imported fuel or locks in domestic electricity for 30 to 40 years.

Why does the world need an energy transition?

- Cleaner air.** The WHO estimates that 99 percent of the world's people breathe air above its guideline limits. Clean generation and electric transport cut that directly.
- Lower bills.** Rooftop solar cuts household power costs, and over its life an electric vehicle is cheaper to run than a petrol car, even if charging is less convenient today.
- Access.** About 655 million people still had no electricity in 2024, over 560 million of them in sub-Saharan Africa. Cheap solar and storage are the fastest route to them.
- National resilience.** Every war and oil price spike makes importing countries poorer. Moving to domestic electricity, while avoiding a new dependence on one supplier, is now a national mission.

Why policy moves in zigzags

Policymakers in a technical sector hear from dozens of companies, many arguing for what suits them rather than the country. Telling the difference takes experience, so policy rarely moves in a straight line: two steps forward, one step back, sometimes sideways. The swings in US policy between 2022 and 2025 show how far that can go.

The prize

Energy is the largest investment sector in the world, and clean energy now takes more of it than fossil fuels. Whoever develops a new energy source that is cheap, scalable and patentable will create one of the largest fortunes in history.

Where the Business Opportunities Are, and Our View

Generation is crowded. The next returns sit around the power plant, not inside it.

Where a newcomer can still earn a return

OPPORTUNITY	WHY IT IS OPEN	CAPITAL NEED
Components outside China	Western and Indian buyers need non-Chinese supply, viable only behind durable trade protection	High
EPC, construction and maintenance	Close to 700 GW a year to build and run, and few independent contractors at institutional standard	Medium
Grid equipment and services	About 2,500 GW of projects wait for connections worldwide	Medium to high
Grid and storage software	Forecasting, battery dispatch and demand response, with no dominant player yet	Low
Storage projects	112 GW added in 2025, up 48 percent, as midday surplus spreads	Medium
Clean power for data centres	Demand set to more than double to about 945 TWh by 2030	High
Energy access in Africa	655 million people without power, yet Africa took 1.6 percent of 2025 additions	Medium
Sizing the global pool	In 2026 the world is expected to invest about USD 665 billion in renewable power, USD 550 billion in grids and more than USD 100 billion in battery storage, over USD 1.3 trillion in total.	

Three policy priorities for governments

- 1 Supply chain.** Protect domestic manufacturing broadly, not through incentives that pick a few winners.
- 2 End use.** Electrify transport, buildings and industry faster.
- 3 Grid.** Build lines and batteries much faster, because everything else depends on them.

YD Advisory View

Underwrite the bottleneck, not the panel. Module and turbine prices are set in China. Grids, storage, land, connections and specialist contracting carry scarcity value in most markets this decade.

Price policy risk explicitly. Manufacturing returns rest on import protection and listing rules that can change. Model the downside without them.

Merchant risk is rising. Midday surplus and curtailment widen the gap between solar-only and firm supply. Storage changes the revenue stack and the valuation.

Cost of capital is the swing factor. A one point move in the financing rate often changes equity returns more than most operating assumptions.

For Gulf-based capital, 2026 showed the region's own export route can close. Clean power, grid and storage assets in import-dependent markets across Asia, Europe and Africa are a natural hedge.

Bottom line. Renewables became a major business opportunity because they got cheap, then became essential. The next phase rewards whoever solves storage, grids and supply chain dependence.



Boutique transaction and strategic advisory. In energy and infrastructure we build the financial case, test the assumptions and structure the capital stack.

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Figures reflect published data available at 1 October 2026 and may be revised. YD calculations, estimates and views are labelled where used. Full source list available on request.



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